

Lasai Barreñada

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PROFESSIONAL EXPERIENCE

KU Leuven

PhD Researcher in clinical prediction models (Leuven) 10/2022-Current

Memorial Sloan Kettering Cancer Center

Visiting PhD Researcher (New York) 02/2025-06/2025

Julius center - UMC Utrecht

Visiting PhD Researcher (Utrecht) 10/2024-02/2025

Basque Center for Applied Mathematics (BCAM)

Research Technician in Machine Learning (Bilbao) 12/2021-09/2022

Spanish National Statistical Office (INE Spain)

Postgraduate Internship (Madrid) 10/2020-09/2021

EDUCATION

KU Leuven, PhD in Biomedical Sciences 2022-2026 (expected)

Complutense University of Madrid (UCM), European Master in Official Statistics (EMOS) 2019-2021

University of Granada, B.A. Marketing 2015-2019

PUBLICATIONS

1. Van Calster, B. *et al.* Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: Overview and guidance. *The Lancet Digital Health* **0**, (2025).
2. Barreñada, L. *et al.* Head-to-head comparison of the RMI and ADNEX models to estimate the risk of ovarian malignancy: A systematic review and meta-analysis of external validation studies. *BMJ Open* **15**, e104141 (2025).
3. Zegarra, R. R. *et al.* Standardization of the sonographic measurement of the head to perineum distance: A pilot study comparing techniques with and without perineal compression. *European Journal of Obstetrics and Gynecology and Reproductive Biology* **312**, (2025).
4. Moro, F. *et al.* External validation of ultrasound-based models for differentiating between benign and malignant adnexal masses: A nationwide prospective multicenter study (IOTA phase 6). *American Journal of Obstetrics and Gynecology* **S0002-9378(25)00477-6** (2025) doi:[10.1016/j.ajog.2025.07.017](https://doi.org/10.1016/j.ajog.2025.07.017).
5. Geysels, A. *et al.* Artificial intelligence applied to ultrasound diagnosis of pelvic gynecological tumors: A systematic review and meta-analysis. *Gynecologic and Obstetric Investigation* (2025) doi:[10.1159/000545850](https://doi.org/10.1159/000545850).
6. Barreñada, L., Campo, B. D. C., Wynants, L. & Calster, B. V. Clustered flexible calibration plots for binary outcomes using random effects modeling. (2025) doi:[10.48550/arXiv.2503.08389](https://doi.org/10.48550/arXiv.2503.08389).
7. Martínez-García, M., García-Gutierrez, S., Barreñada, L., Inza, I. & Lozano, J. A. Extending the learning using privileged information paradigm to logistic regression. *Neurocomputing* **615**, 128869 (2025).
8. Van Calster, B. *et al.* Performance evaluation of predictive AI models to support medical decisions: Overview and guidance. (2024) doi:[10.48550/arXiv.2412.10288](https://doi.org/10.48550/arXiv.2412.10288).

9. Barreñada, L., Dhiman, P., Timmerman, D., Boulesteix, A.-L. & Van Calster, B. [Understanding overfitting in random forest for probability estimation: A visualization and simulation study](#). *Diagnostic and Prognostic Research* **8**, 14 (2024).
10. Barreñada, L. *et al.* [ADNEX risk prediction model for diagnosis of ovarian cancer: Systematic review and meta-analysis of external validation studies](#). *bmjmed* **3**, e000817 (2024).
11. Martínez-García, M. *et al.* *Clinical Severity Prediction of COVID-19 Admitted Patients in Spain: SEMI and REDISSEC Cohorts*. <http://medrxiv.org/lookup/doi/10.1101/2023.02.08.23285589> (2023) doi:10.1101/2023.02.08.23285589.
12. Barreñada, L., Gálvez Sainz de Cueto, J. C. & Fernández Calatrava, J. [Timeliness reduction on industrial turnover index based on machine learning algorithms](#). *Statistical Journal of the IAOs* **38**, 1195–1205 (2022).

PEER REVIEW

Conducted various peer reviews for **Statistics in Medicine**, **Ultrasound in Obstetrics and Gynecology**, **International Journal of Gynecological Cancer**, **Heliyon**, **npj precision oncology**, **Nature Communications medicine** and **Diagnostic and prognostic research**

CONFERENCE PRESENTATIONS

Updating the ADNEX model for surgically and conservatively managed patients (Talk) , ISUOG, Cancun	09/2025
Combining calibration plots (Talk) , MEMTAB, Birmingham	04/2025
Problem of Individual Risk Estimation for Health AI (Talk) , ENAR, New Orleans	03/2025
Head-to-head comparisons of RMI and ADNEX models (Talk) , ISUOG, Budapest	09/2024
Multicenter flexible calibration curves using random effects (Poster) , ISCB, Thessaloniki	07/2024
Drawing multicenter calibration curves (Talk) , SEB, Barcelona	02/2024
ADNEX model systematic review and meta-analysis (Talk) , ISUOG, Seoul	10/2023
Do we understand random forests? (Poster) , ISCB, Milan	08/2023
Do we understand random forests? (Poster) , YSM, Leicester	07/2023

AWARDS & PRIZES

Young Investigator Award in Gynaecology , issued by ISUOG	2023
Young Statisticians Prize (YSP) , 2 nd , issued by IAOS	2022

SKILLS

Clinical Prediction Models
 Meta-analysis
 Machine Learning
 Statistical Analysis
 Data Science
 Biomedical Research
 Programming: **R**, **Python**

PROFESSIONAL MEMBERSHIPS

STRATOS early career adjunct member

International Society of Ultrasound in Obstetrics and Gynecology (ISUOG)
International Ovarian Tumour Analysis group (IOTA)
International Biometric Society (IBS)
International Society For Clinical Biostatistics (ISCB)

LANGUAGES

Spanish: Native

Basque: Native

English: Full Working Proficiency

French: Full Working Proficiency